

Supplementary Materials

Morphology-Aware Neuro-Symbolic Relation Extraction for Manipuri Using Relational Graph Neural Networks and Tsetlin Machines

Table 1: Representative correctly classified relation instances and their explicit Tsetlin Machine evidence. The examples are taken from the model evidence output. H-T denotes head-before-tail and T-H denotes tail-before-head.

Relation	Entity Pair	Entity-Type Pair	Order	Dist	Morphological Evidence
AFFECTS	terrorism → India	Event → Location	H-T	≤ 2	–
AFFECTS	violence → people	Event → Group	H-T	> 5	–
AFFECTS	women police → rally	Organization → Event	T-H	≤ 5	–
AFFILIATED_WITH	MLA → party	Title → Organization	T-H	≤ 2	Genitive: BETWEEN_HAS_GENITIVE, TAIL_HAS_GENITIVE
AFFILIATED_WITH	minister → party	Title → Organization	T-H	≤ 5	Genitive + Locative
AFFILIATED_WITH	SPF → government	Organization → Organization	H-T	≤ 2	Genitive
ASSOCIATED_WITH	government → Delhi	Organization → Location	H-T	> 5	Genitive + Locative
ASSOCIATED_WITH	nematode → white tip	Disease/Pest → Disease/Pest	T-H	≤ 5	–
ASSOCIATED_WITH	terrorism → Jaipur	Event/Concept → Location	T-H	≤ 5	–
HAS_VALUE	ID → email address	Identifier → Email-ID	H-T	≤ 2	–
HAS_VALUE	ID → email address	Identifier → Email-ID	H-T	≤ 2	–
HAS_VALUE	Lungmila → 60	Person → Age	H-T	≤ 2	–
HOLDS_TITLE	Ibobi → Chief Minister	Person → Title	T-H	≤ 2	Locative
HOLDS_TITLE	O. Ibobi → Chief Minister	Person → Title	T-H	≤ 2	–
HOLDS_TITLE	Gaikhangam → President	Person → Title	T-H	≤ 2	–
LOCATED_IN	seed → container	Seed → Container	H-T	≤ 5	–

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Relation	Entity Pair	Entity-Type Pair	Order	Distance	Morphological Evidence
LOCATED_IN	bomb → Jaipur	Weapon/Event → Location	T-H	≤ 2	–
LOCATED_IN	bombs → Hanuman Temple	Weapon/Event → Facility	H-T	≤ 5	–
MIXED_WITH	medicine → water	Chemical/Treatment → Resource	T-H	> 5	–
MIXED_WITH	medicine → water	Chemical/Treatment → Resource	H-T	≤ 2	–
MIXED_WITH	medicine → water	Chemical/Treatment → Resource	H-T	> 5	–
OPPOSES	opposition → government	Organization → Organization	T-H	≤ 2	–
OPPOSES	opposition → government	Organization → Organization	H-T	> 5	Copula: BETWEEN_HAS_COPULA
OPPOSES	opposition parties → government	Organization → Organization	T-H	> 5	Conjunctive + Locative
PARTICIPATES_IN	group → rally	Group → Event	H-T	> 5	–
PARTICIPATES_IN	Meira Paibi → rally	Group → Event	H-T	≤ 5	–
PARTICIPATES_IN	groups → rally	Group → Event	T-H	> 5	–
PART_OF	party organization → party	Organization → Organization	T-H	> 5	Genitive + Locative
PART_OF	Shakarpur → New Delhi	Location → Location	T-H	≤ 2	–
PART_OF	RDX → bomb	Explosive Material → Weapon/Event	H-T	> 5	–
TREATS	treatment → seeds	Chemical/Treatment → Seed	H-T	≤ 5	–
TREATS	treatment → seed	Chemical/Treatment → Seed	T-H	≤ 2	–
TREATS	treatment → seed	Chemical/Treatment → Seed	T-H	> 5	–
USES	police → tear gas	Organization → Control Measure	H-T	≤ 2	–
USES	police → mock bomb	Organization → Control Measure	H-T	≤ 5	–

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Relation	Entity Pair	Entity-Type Pair	Order	Dis- tance	Morphological Evidence
USES	police tear gas	→ Organization → Control Measure	H-T	> 5	–